

**Package:** In ROS 2, a package is simply a folder that organizes specific files and functionalities related to a robotic component.

→ Organization: It keeps all files for a specific job - such as localization or sensor data in one place.

→ Contents: It contains source code, configuration files, readme files and custom message types needed for that functionality.

→ Modularity: This structure allows you to maintain, debug and reuse specific code components across different robotics projects efficiently.

| ros2 pkg list | list installed <del>ROS</del> ROS2 packages |
|---------------|---------------------------------------------|
|               |                                             |

`ros2 pkg create --build-type ament_cmake --license Apache 2.0 ros2_fundamentals`  
(package name)

- `ros2 pkg create`: command to create a package  
→ commonly used for C++

→ --build-type ament\_cmake: option + its value to specify build type

→ --license Apache 2.0: option + its value to specify license



```
colcon build --packages-select ros2_fundamentals  
(package name)
```

Command to build one specific ROS2 package instead of the entire workspace.

★ This is highly efficient when you want to compile and test changes for a single project without waiting for the entire workspace to ~~rebuild~~ rebuild.

Some external packages to enhance ROS2 development

- ↳ ROS GZ (Gazebo): control/controller packages for simulation.
- ↳ Joint state GVI: testing & debugging joint movements.
- ↳ Moveit: motion planning.
- ↳ xacro: working with URDF/robot description files.
- ↳ terminator: terminal emulator that allows split window.
- ↳ libserial-dev: serial communication projects.
- ↳ Python3-pip: to manage python dependencies.